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1 Abstract

This project focuses on developing a predictive football betting model that integrates team ratings calculated using Elo method with logistic regression to optimize betting strategies. Elo ratings, a widely used method in sports ranking, provide a measure of team performance based on historical match results. This project enhances traditional Elo ratings by incorporating logistic regression calibration, which refines the prediction of match outcomes and provides a more accurate representation of probabilities. The approach utilizes historical english premier league match data from 2014-2024 to train Elo ratings and then employs logistic regression to calibrate the predicted probabilities for the ongoing 2024-25 season.

2 Introduction

Football betting is a multi-billion dollar industry, attracting professional algorithmic strategies. The goal of sports betting is to predict the outcome of matches in a way that generates consistent profit. However, the complexity of match outcomes, influenced by numerous variables such as each player, team strength, form, injuries, and external factors, makes it difficult to predict. To optimize betting decisions, it is essential to have an effective model that can accurately predict match outcomes and maximize returns over time.

One of the most established methods for quantifying strength in sports is the Elo rating system, which was originally developed by Arpad Elo to rank chess players. The Elo system evaluates teams based on their performance in head-to-head matches, adjusting ratings depending on match outcomes and the relative strength of opponents. This approach has been widely adopted in various sports, including football, as it provides a objective measure of a team's strength and current performance. However, while Elo ratings are effective at ranking teams, they require additional methods to be leveraged for betting purposes, as they do not directly provide the probabilities of winning a match.

This project aims to combine the Elo rating system with logistic regression to develop a predictive football betting model. The model uses the Elo ratings to produce more accurate predictions of match outcomes and provide more reliable betting decisions.

3 Dataset Description

The dataset used in this project spans 11 seasons of football data, covering the years from 2014-2024, along with data from the ongoing 2024-2025 season. This dataset includes detailed match outcomes, team performance statistics, and betting odds from multiple bookmakers. It provides relevant data for developing a predictive football betting model. The data offers comprehensive insights into team performance, betting odds, and match results, making it ideal for optimizing football betting strategies.

3.1 Key Columns in the Dataset

The columns relevant to our analysis are outlined below:

- *Date*: The date when the match was played, essential for time-based analysis.
- *HomeTeam*: The team playing at home, used for Elo rating calculations and predictions.
- *AwayTeam*: The team playing away, used for Elo rating calculations and predictions.
- *FTHG (Full Time Home Goals)*: The number of goals scored by the home team during the match, used to update Elo ratings.
- *FTAG (Full Time Away Goals)*: The number of goals scored by the away team during the match, used to update Elo ratings.
- *FTR (Full Time Result)*: The outcome of the match, denoted as H (Home Win), A (Away Win), or D (Draw), used for outcome-based betting decisions.
- *B365H, B365D, B365A*: The odds from Bet365 for *home win*, *draw*, and *away win* outcomes, used to calculate implied probabilities.
- *MaxH, MaxD, MaxA*: The maximum odds available for *home win*, *draw*, and *away win* outcomes, used to identify the best betting options across bookmakers.
- *AvgH, AvgD, AvgA*: The average odds for *home win*, *draw*, and *away win* outcomes, which help assess general market tendencies and calculate expected outcomes.

The dataset is split into two distinct parts:

- **Training Data**: Includes matches from seasons 2014-15 to 2023-24. This is used to calculate the Elo ratings and build the predictive model using logistic regression for match outcomes.
- **Testing Data**: Includes matches from the 2024-2025 season and is used to evaluate the performance of our betting model, assess profitability, and validate the model.

3.2 Data exploration

We now look at the basic descriptive data of 3800 matches from 10 seasons.

Descriptive Statistics for Goals

Statistic	Home Goals (FTHG)	Away Goals (FTAG)
Mean	1.5479	1.2453
Median	1.0	1.0
Std Dev	1.3204	1.2025
Min	0.0	0.0
Max	9.0	9.0

(1)

Match Result Distribution (FTR)

Result	Count
Home Win (H)	1708
Draw (D)	886
Away Win (A)	1206

(2)

Correlation Between Match Statistics

$$\begin{bmatrix} & \text{FTHG} & \text{FTAG} & \text{HS} & \text{AS} & \text{HC} & \text{AC} \\ \text{FTHG} & 1.000 & -0.127 & 0.316 & -0.173 & 0.043 & -0.087 \\ \text{FTAG} & -0.127 & 1.000 & -0.157 & 0.350 & -0.095 & 0.057 \\ \text{HS} & 0.316 & -0.157 & 1.000 & -0.421 & 0.547 & -0.350 \\ \text{AS} & -0.173 & 0.350 & -0.421 & 1.000 & -0.348 & 0.538 \\ \text{HC} & 0.043 & -0.095 & 0.547 & -0.348 & 1.000 & -0.305 \\ \text{AC} & -0.087 & 0.057 & -0.350 & 0.538 & -0.305 & 1.000 \end{bmatrix}$$

Goals by Match Result (FTR)

Result	**Mean Home Goals**	**Median**	**Mean Away Goals**	**Median**
H	2.499	2.0	0.552	0.0
D	1.029	1.0	1.029	1.0
A	0.582	0.0	2.386	2.0

The following data and plots suggest a normal distribution of goal difference around mean = 0. This suggest that a betting strategy that only bets on Home team will also have 50 percent win rate.

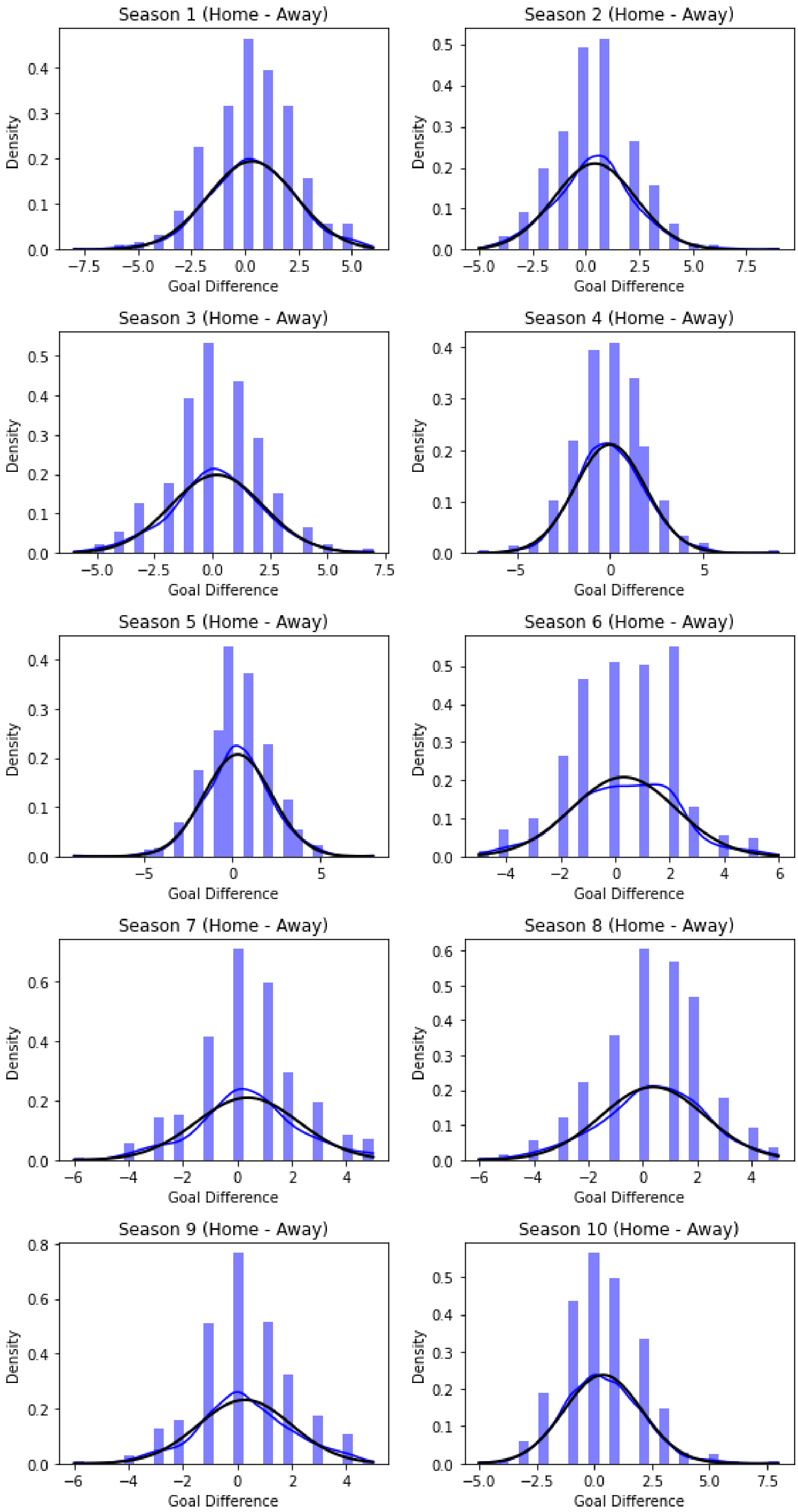

Figure 3.1: Goal difference (Home goals - Away goals) plot for the training data

4 Methodology

This section develops a statistical betting strategy for football matches by integrating an Elo rating system with a calibrated logistic regression model to estimate match outcome probabilities. These probabilities are then compared against bookmaker odds to identify value bets. Below, we provide a explanation of our approach.

4.1 Elo Rating System for Team Strength Estimation

The *Elo rating system* is used to update the strength of football teams after each match based on the outcome. The Elo rating system assigns each team a numerical score, and the relative difference in scores between two teams is used to predict the outcome of their match.

Initially, all teams are assigned an equal starting Elo rating. The Elo rating system is a way of calculating the relative skill levels of teams based on match outcomes, with the goal of updating each team's rating after each match.

Elo Rating Initialization: At the beginning, we initialize each team with the same base rating $R_0 = 1000$. This is done for all teams participating in the dataset. The teams' Elo ratings are then updated after each match.

$$R_{\text{team}} = 1000 \quad \text{for all teams.}$$

4.1.1 Expected Probability of Outcome

Given two teams A and B, the probability that team A wins (i.e., the home team wins) can be expressed as the expected probability E_A of team A winning. This can be formulated using the difference of elo ratings for this probability is given by:

$$E_A = \frac{1}{1 + 10^{\frac{R_B - (R_A + H)}{400}}}$$

Where:

- R_A and R_B are the Elo ratings of teams A and B, respectively.
- H is the home advantage constant, typically a fixed value of around 75 Elo points, which adjusts the home team's advantage.
- E_A is the expected probability that the home team A wins the match.

The expected probability for the away team B, E_B, is simply:

$$E_B = 1 - E_A$$

This formula reflects the principle that if the home team is more likely to win, the away team's probability of winning will be lower.

4.1.2 Actual Outcome

After the match, the actual outcome is determined based on the final scores. The actual outcome S_A for team A and S_B for team B is defined as follows:

$$S_A = \begin{cases} 1, & \text{if team A wins} \\ 0.5, & \text{if it's a draw} \\ 0, & \text{if team A loses} \end{cases}$$

Similarly for S_B:

$$S_B = 1 - S_A$$

This reflects that the result of a match can be win, draw, or loss, with corresponding scores of 1, 0.5, or 0, respectively.
Once the logistic regression model is trained, it provides calibrated probabilities $\hat{p}_A$, which are often more accurate than the raw Elo probabilities, accounting for real-world data deviations and trends.

4.1.3 Elo Rating Update

Once the match outcome is determined, we update the Elo ratings for both teams. The updated Elo ratings R'_A and R'_B are computed as follows:

$$R'_A = R_A + K \cdot (S_A - E_A)$$

$$R'_B = R_B + K \cdot (S_B - E_B)$$

Where: - K is the Elo update factor, controlling the sensitivity of the system to match results. A higher K value leads to faster rating adjustments. Typically, K is set between 20 and 100. - S_A and S_B are the actual scores, and E_A and E_B are the expected scores based on the teams' Elo ratings.
The ratings are updated after each match to reflect the current relative strength of the teams.

4.2 Logistic Regression Calibration

Although the Elo rating system provides a theoretical estimate of match outcomes based on team rating differences, these probabilities may not always be perfectly aligned with real-world match results. To

Algorithm 1 Elo Rating Update Algorithm

Input:
Team ratings $R_{\mathrm{h}}, R_{\mathrm{a}}$
Home advantage H
Actual match result $S_{\mathrm{h}}, S_{\mathrm{a}}$
Elo constant K
Step 1: Calculate the expected probability of the home team winning

$$E_{\mathrm{h}} = \frac{1}{1 + 10^{\frac{R_{\mathrm{a}} - (R_{\mathrm{h}} + H)}{400}}}$$

$$E_{\mathrm{a}} = 1 - E_{\mathrm{h}}$$

Step 2: Determine the actual outcome of the match
if goals_home $>$ goals_away **then**
  $S_{\mathrm{h}} = 1$, $S_{\mathrm{a}} = 0$
else if goals_home $<$ goals_away **then**
  $S_{\mathrm{h}} = 0$, $S_{\mathrm{a}} = 1$
else
  $S_{\mathrm{h}} = 0.5$, $S_{\mathrm{a}} = 0.5$
end if
Step 3: Update the Elo ratings

$$R'_{\mathrm{h}} = R_{\mathrm{h}} + K \cdot (S_{\mathrm{h}} - E_{\mathrm{h}})$$

$$R'_{\mathrm{a}} = R_{\mathrm{a}} + K \cdot (S_{\mathrm{a}} - E_{\mathrm{a}})$$

Step 4: Return updated ratings
Output: Updated team ratings $R'_{\mathrm{h}}, R'_{\mathrm{a}}$ =0

address this, we apply *logistic regression* to calibrate the predicted probabilities derived from the Elo system.

4.2.1 Logistic Regression Model

We use logistic regression to calibrate the raw probabilities produced by the Elo model. Given the Elo rating difference between teams A and B, denoted as x_i, the logistic regression model is used to predict the probability $\hat{p}_A$ that the home team A wins. Let the Elo difference x_i be:

$$x_i = R_A - R_B$$

The logistic regression model is defined as:

$$\mathbb{P}(Y_i = 1 | x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

Where:

- $Y_i = 1$ if the home team wins match i, and $Y_i = 0$ if the home team loses (binary outcome).
- β_0 and β_1 are the parameters of the logistic regression model, which are learned via maximum likelihood estimation. β_1 represents the impact of the Elo rating difference on the probability of a

home win.

Once the logistic regression model is trained, it provides calibrated probabilities $\hat{p}_A$, which are often more accurate than the raw Elo probabilities, accounting for real-world data deviations and trends.

4.2.2 Calibrated Probability

After training, the logistic regression model outputs a calibrated probability $\hat{p}_A$ that the home team wins, which is used for subsequent predictions. This probability is expected to be more reliable than the raw Elo probability because it incorporates historical deviations and adjustments based on empirical data.

4.3 Value Betting Strategy

The heart of the betting strategy lies in identifying value bets, where the model's predicted probability of an outcome exceeds the implied probability based on the bookmaker's odds. This allows us to place bets on outcomes that are expected to be profitable in the long run.

4.3.1 Implied Probability from Odds

Given the decimal odds o from a bookmaker, the implied probability π of an event is given by:

$$\pi = \frac{1}{o}$$

For the home win outcome, the edge is calculated as the difference between the predicted probability from the calibrated model $\hat{p}_A$ and the implied probability π_{home}:

$$\text{Edge} = \hat{p}_{\text{home}} - \pi_{\text{home}}$$

This edge represents the difference between the model's prediction and the bookmaker's implied probability. A positive edge indicates a potential value bet.

4.3.2 Betting Decision

A bet is placed if the calculated edge exceeds a threshold δ (e.g., 0.02 or 2

$$\text{Edge} > \delta$$

If the edge is greater than the threshold, the model places a bet on the home team; otherwise, the model bets on the away team, or skips the bet entirely.

4.3.3 Profit Calculation

Once a bet is placed, the profit is calculated based on the outcome. If the bet wins, the profit is:

$$\text{Profit} = (o - 1) \cdot B$$

Where:

- o is the decimal odds (e.g., 1.90, 2.50).
- B is the flat bet amount (e.g., 100).

If the bet loses, the loss is:

$$\text{Profit} = -B$$

The bankroll is updated after each bet, and it is tracked over time:

$$\text{Bankroll}_{t+1} = \text{Bankroll}_t + \text{Profit}_t$$

4.4 Smoothing Elo Ratings

To reduce volatility in Elo rating updates, a *smoothed Elo rating* $\bar{R}_i^{(t)}$ is computed as a rolling average of the last n matches:

$$\bar{R}_i^{(t)} = \frac{1}{n} \sum_{j=t-n+1}^{t} R_i^{(j)}$$

Where:

- n is the smoothing window, typically set between 5 and 15 matches.
- The smoothed rating is used in place of the immediate Elo rating when calculating the predicted probability.

4.5 Evaluation Metrics

The model's performance is evaluated using two commonly used metrics: Brier score and log loss, both of which quantify the accuracy of the predicted probabilities.

4.5.1 Brier Score

The Brier score is a measure of the mean squared difference between the predicted probabilities and the actual outcomes:

$$BS = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2$$

Where:

- p_i is the predicted probability for match i,
- o_i is the actual outcome (1 for win, 0 for loss).

4.5.2 Log Loss

The log loss is calculated as:

$$LL = -\frac{1}{N}\sum_{i=1}^{N}\left[o_i \log(p_i) + (1 - o_i)\log(1 - p_i)\right]$$

Where:

- p_i is the predicted probability,
- o_i is the actual outcome.

5 Result

The results of the simulation were obtained through a grid search process, which explored different combinations of the key parameters, including the Elo rating update factor K, the home advantage constant HomeAdv, and the edge threshold EdgeThresh. The most optimal combination was found with $K = 38$, HomeAdv $= 25$, and EdgeThresh $= 0.11$.

5.1 Evaluation Metrics

- **Brier Score:** The model's Brier score was calculated as 0.2176, which reflects the mean squared difference between predicted probabilities and actual outcomes. A lower Brier score indicates better performance, suggesting that the model's predicted probabilities are reasonably close to the actual outcomes.
- **Log Loss:** The log loss for the simulation was 0.6249, indicating the accuracy of the predicted probabilities. While this value suggests room for improvement, it also indicates that the model is able to make reasonably confident predictions about match outcomes.

5.2 Betting Simulation

- **Total Bets:** The model placed 277 bets across the 2024-2025 season, based on the predicted edge values and the bookmaker odds.

- **Win Rate:** The win rate for the betting strategy was 46.21%, meaning that just under half of the bets placed resulted in a win. This is a promising result, given that a typical win rate for a profitable betting strategy is often around 50%.
- **Total Profit:** The simulation yielded a total profit of 5139 on a bankroll of 10000, which is the net gain after accounting for all bets placed.
- **Return on Investment (ROI):** The ROI for the betting strategy was 18.76%, indicating that for every 100 bet, the model returned 18.76 in profit. This ROI suggests that the model's betting strategy is profitable over time.

5.3 Top Teams by Final Elo Rating

The Elo ratings were updated after each match, with the final ratings for the top 10 teams in the league as of the end of the season being as follows:

Rank	Team	Final Elo Rating
1	Liverpool	1336.94
2	Arsenal	1275.60
3	Man City	1218.80
4	Aston Villa	1149.61
5	Newcastle	1141.25
6	Nott'm Forest	1134.88
7	Chelsea	1122.56
8	Bournemouth	1120.28
9	Brighton	1119.87
10	Fulham	1111.11

We get the following elo plots over time using our dataset including the test data.

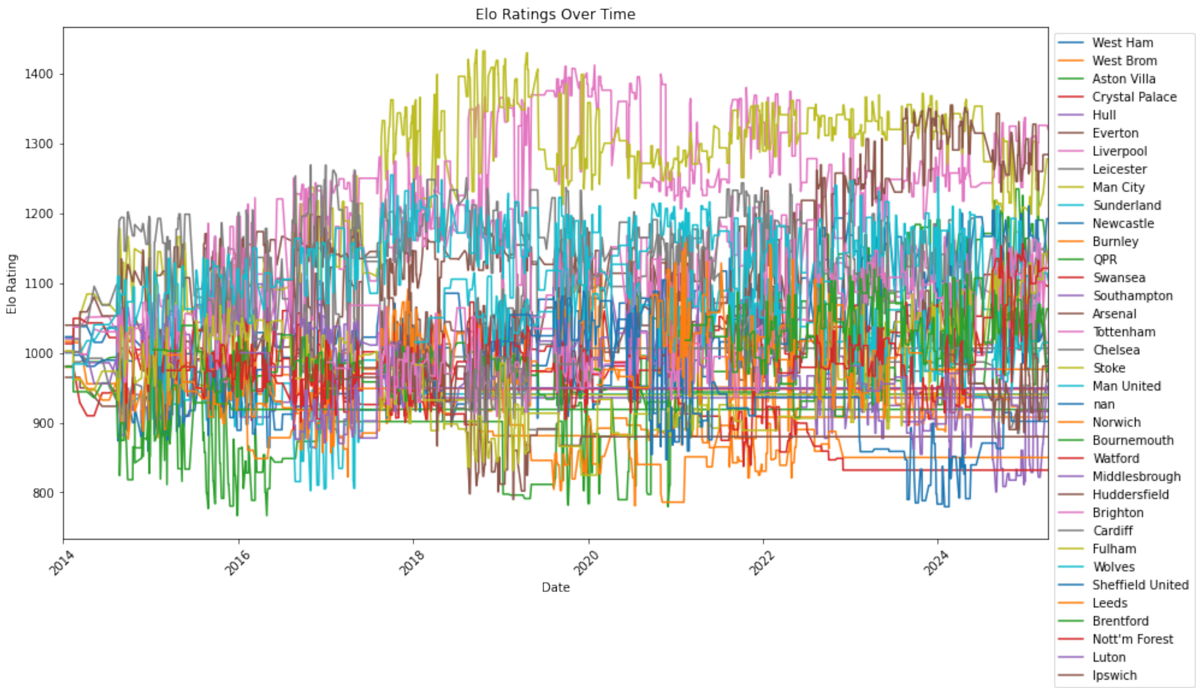

Figure 5.1: ELO ratings of teams over time

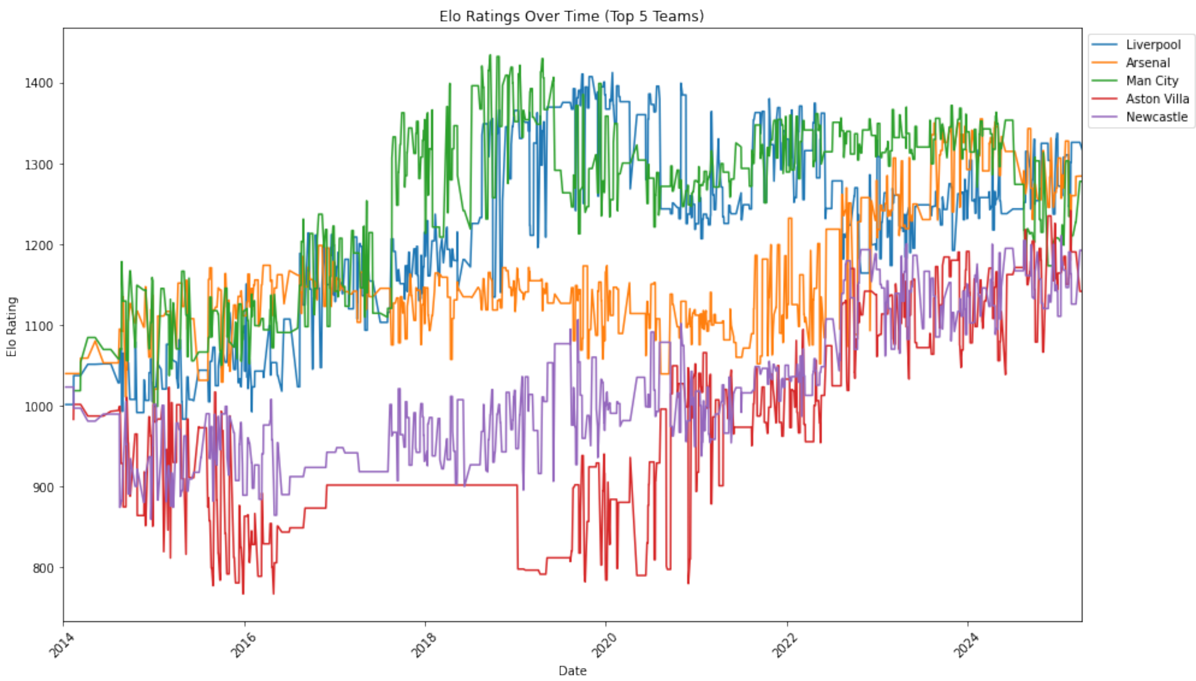

Figure 5.2: top 5 teams over time based on elo ratings

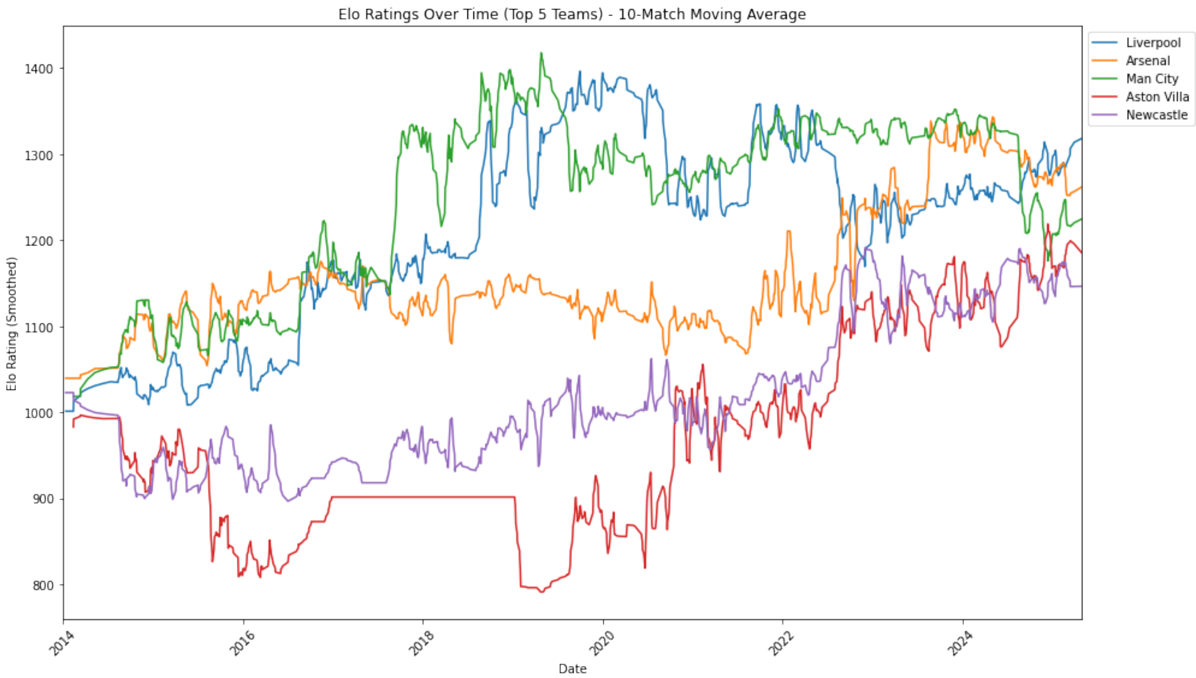

Figure 5.3: Smooth elo using moving averages of window 10

6 Conclusion

The model demonstrated a solid performance in terms of profitability, with a strong ROI of 18.76%. The Elo ratings successfully captured the relative strengths of the teams, and the logistic regression calibration further improved the prediction accuracy. The Brier score and log loss metrics suggest that there is room for model improvement, particularly in refining the predicted probabilities. However, the simulation results provide a promising foundation for using the Elo rating system combined with logistic regression as a football betting strategy.